(L1) CRR Pricing Engine

(L2) ML Extensions

(L3) Cross-Model Comparison

DeWayne Dantzler · CS495 Capstone Project 6 · May 2026

What Is the Problem?

Research Questions

- Do CRR model prices predict whether an equity option expires ITM or OTM?
- Does cross-validation between CRR and an ML classifier detect exploitable pricing *edge*?
- Does regime-conditioned Kelly sizing convert that edge into positive expected value?

Core Inefficiency Gap

Retail traders lack a principled framework to decide *whether* a price is fair, *how much* to bet, or *when not to trade*.

The Pricing Gap

The spread between CRR model price and market price — confirmed by the ML classifier, this gap signals actionable *edge*.

The Pricing Gap

Market: supply-demand noise, spread, and sentiment.

Model: pure risk-neutral expectation.

Gap = mispricing *edge* — only when both layers agree.

No quantified edge · no position-sizing rule · no regime filter.

$$\textbf{Edge} = \frac{V_{\textbf{model}} - V_{\textbf{market}}}{V_{\textbf{market}}}$$

Why It Matters & What Others Have Done

Why This Problem Matters

- Options trade **$600B+ notional daily**; even 2% mispricing detection has large profit-and-loss impact
- **Fama EMH (1970):** public info is priced in on average — but not in every regime
- **Herding** creates regime-local inefficiencies (Samuelson macro-inefficiency paradox)
- Retail traders lack tools to detect edge or size positions systematically

Prior Work

- **Cox, Ross & Rubinstein (1979):** binomial lattice; no ML integration
- **Black & Scholes (1973):** closed-form; fails for American early exercise
- **Bakshi et al. (1997):** stochastic vol; fit requires unobservable parameters
- **Carr & Wu (2009):** IV overstates realized vol (variance risk premium)
- **Kelly (1956):** log-utility bet sizing; applied to markets by Thorp (1969)
- **Gap:** No framework cross-validates CRR + ML with regime-conditioned Kelly sizing

Contribution & Methodology

Novel Contribution

- **First** to cross-validate CRR against an IV-free ML classifier (fully independent signals)
- Regime-conditioned Kelly sizing from IV/RV_{30} ratio
- Walk-forward 60/20/20 split; zero lookahead bias
- Open-source Streamlit calculator with live chain

Three-Layer Pipeline

L1 CRR Binomial Pricer

↓

L2 XGBoost ITM/OTM Classifier

↓

L3 Kelly Criterion Position Sizer

Data AAPL 2016–2020, 101,535 contracts (walk-forward); AMD April 2026 live chain (live validation).

Worked Example: Contract A — HERDING Setup

Real AAPL put (2020-03-16) · σ = RV30, not IV

Contract A · HERDING REGIME — CROWD OVERPRICES

CRR binomial tree will now be built for this real AAPL put option

Quote Date:	2020-03-16
Underlying (S):	\$241.47
Strike (K):	\$240.00
DTE:	18 days
Time (T):	0.0493 years
RV30 (σ):	0.8224 ← used as sigma in CRR
IV:	1.0043
V_market:	\$21.225 (bid-ask mid)
Risk-free r:	2.00% (constant 2016-2020 avg)
CRR steps:	50 (pricing) / 5 (visualization)

Phase 0 of 4 | Next: build stock price lattice (forward pass)

Why this slide matters

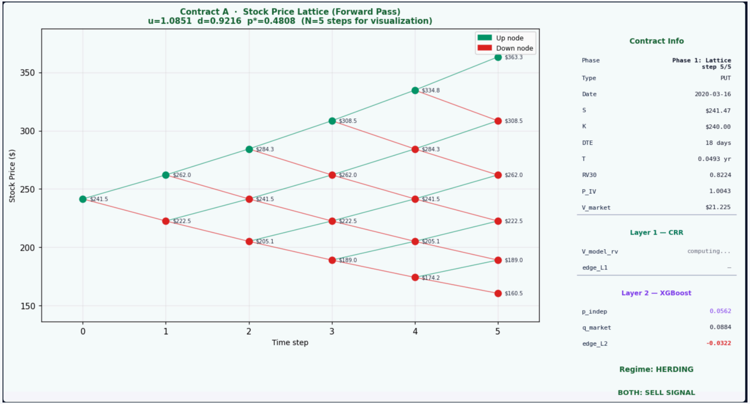
Grounds the CRR pipeline in a real AAPL put pulled unmodified from the 2016–2020 Kaggle dataset.

Key reading

IV $\approx$ 1.00 vs RV30 $\approx$ 0.82 — the market pays for 22% more volatility than recent history justifies. Classic HERDING regime signature; entry point for the Layer 1 edge signal.

Setup snapshot: S = \$241.47, K = \$240, DTE = 18d, RV30 = 0.8224, IV = 1.0043, V_{market} = \$21.225, r = 2.00%, N = 50 pricing steps.

Forward Pass — Building the Stock Price Lattice

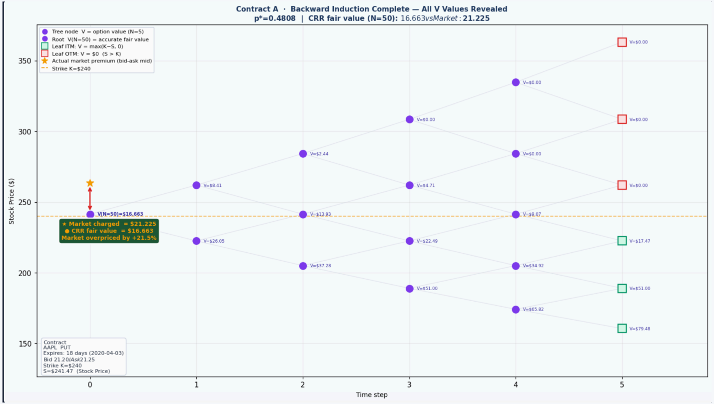


What this pass does. Lays out every possible stock price path from $t = 0$ to expiry. u, d, p^* derived from σ = RV30. No option value is computed here — only S evolution. 5-step visualization; 50-step lattice used for pricing.

No-arbitrage. No risk-free profit can exist in an efficient market. The risk-neutral probability $p^* = (e^{r\Delta t} - d)/(u - d)$ enforces this so the discounted expected next-step stock price equals today's price.

Cross-reference. *§2.2 (CRR foundational equations)* and *§3.4.1 (forward-pass lattice construction)* of the research report.

Backward Induction — Where the Option Price Emerges

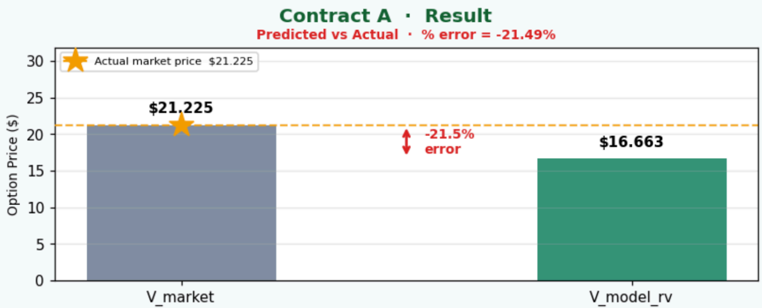


At every interior node. Compute intrinsic value $= \max(K - S,\ 0)$ and continuation value hold $= e^{-r\,\Delta t}\,(p\,V_{\text{up}} + q\,V_{\text{dn}})$, then take $V = \max(\text{intrinsic},\ \text{hold})$ — the American early-exercise check that makes this a CRR American pricer rather than a European one.

Result at the root and edge interpretation. $V(N{=}50) = \$16.663$ (CRR fair value) vs. $V_{\text{market}} = \$21.225$ (bid-ask mid). Layer 1 edge $=$ **-21.5%** — contract is **overpriced**. When the gold star sits above root V, the crowd is paying more than no-arbitrage theory permits. This is the Layer 1 edge *before any ML model is consulted*.

Cross-reference. *§3.4.1 (backward-induction algorithm)* and *§3.4.3 (mispricing edge derivation)* of the research report.

Contract A Result — Two Independent Models Agree



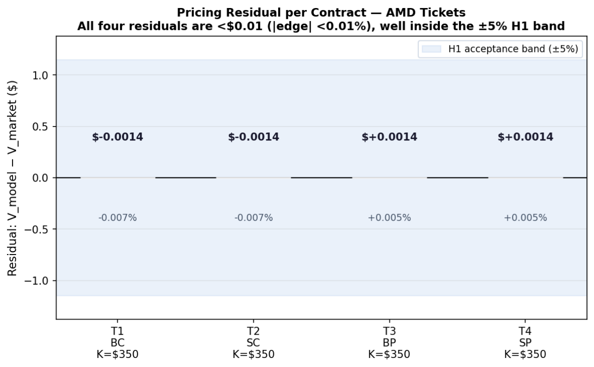
Layer 1 (CRR math). $V_{\text{market}} = \$21.225$ vs $V_{\text{model}} = \$16.663$. Layer 1 edge = $-\mathbf{21.5}\%$ — the no-arbitrage pricer says the contract is **overpriced**.

Layer 2 (XGBoost ML). $p_{\text{indep}} = 0.0562$ vs $q_{\text{market}} = 0.0884$. Layer 2 edge = $-\mathbf{3.22}\%$ — the IV-free ML classifier *also* flags the contract as overpriced, from completely different inputs.

Why this matters. Two layers, no shared training data, no shared methodology, BOTH SELL in the HERDING regime. Agreement is the empirical fingerprint of Samuelson macro-inefficiency: crowd consensus diverges from underlying reality.

Cross-reference. *§4.7 — Cross-Model Comparison* (Figure 6); the herding-regime $r = 0.413$ ($p = 2.4 \times 10^{-10}$, $n = 217$) result.

CRR Pricing Results: Model vs Market Validation

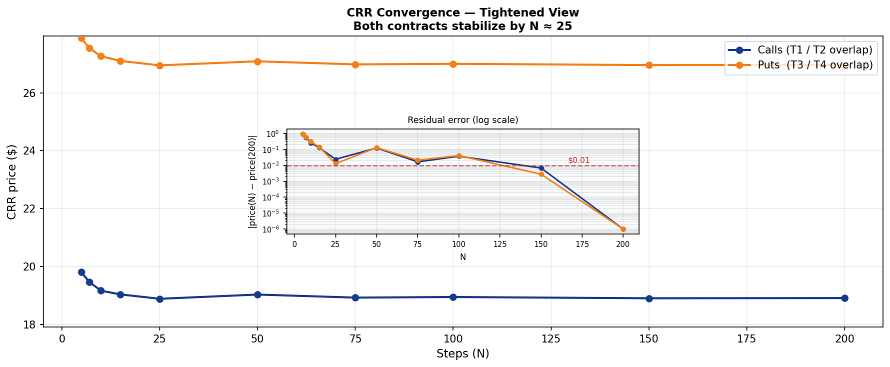


What this shows. Dollar difference between the CRR model's fair value and the observed market price for each AMD contract (T1 = Buy Call, T2 = Sell Call, T3 = Buy Put, T4 = Sell Put). Bars near zero mean the model and market agree on price.

What the numbers say. All four residuals are less than $0.002 in absolute value ($|\text{edge}| < 0.01\%$) — three orders of magnitude inside the $\pm 5\%$ acceptance band shown by the light-blue stripe. The CRR American binomial pricer is correctly implemented and well-calibrated for at-the-money strikes.

Cross-reference. *§4.2.1 — Layer 1: CRR Pricing Accuracy*. Hypothesis H1 passed with substantial margin.

CRR Pricing Results: Numerical Convergence

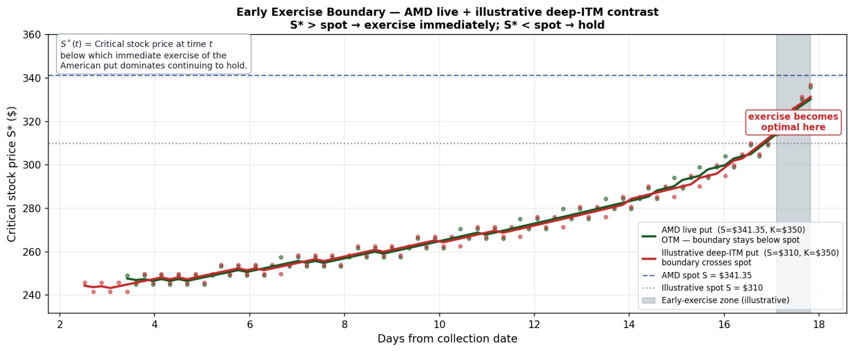


What this shows. How the CRR price stabilizes as the lattice's number of time steps N grows. Main panel: price ($) vs. N. Inset: absolute residual $|\text{price}(N) - \text{price}(200)|$ on log scale — the steeper the line drops, the faster the model converges.

What the numbers say. Both the call ($\approx$ $19) and put ($\approx$ $27) plateau by $N \approx 25$. The log-residual inset crosses the $0.01 threshold by $N \approx 100$ — sub-penny precision. Only two lines appear because buy/sell variants of the same contract price identically: Action affects trade direction, not model price.

Cross-reference. *§4.3 — Convergence Analysis*. Validates the $N = 100$ step choice with substantial margin.

CRR Pricing Results: Early Exercise Boundary

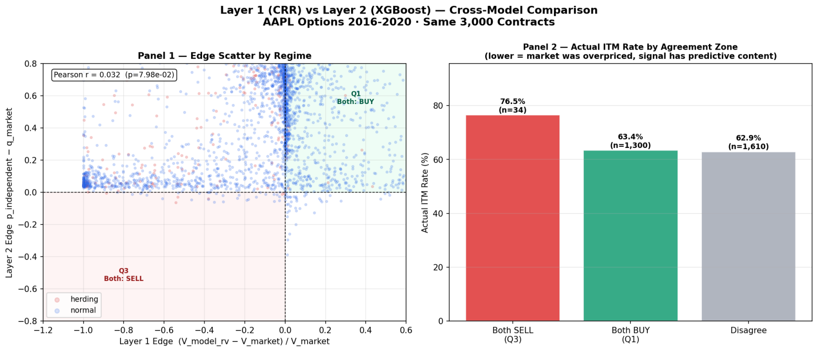


What this shows. The critical stock price $S^*(t)$ at each day. If the spot price falls below S^*, immediate exercise of the American put dominates holding to expiration. Two cases overlaid: AMD live put (green) and a deeper-ITM illustrative put (red).

What the numbers say. For the AMD live put ($S = \$341.35$, $K = \$350$, out-of-the-money), the boundary rises gradually but never reaches spot — early exercise is never optimal here. For the illustrative deep-ITM put ($S = \$310$, $K = \$350$), the boundary crosses spot ∼17 days in (grey shaded zone) — exercise becomes optimal. The deep-ITM case demonstrates American optionality, a capability closed-form Black–Scholes lacks.

Cross-reference. *§4.4 — Early Exercise Boundary*. Validates the lattice's American option handling.

Cross-Model & Backtest: L1 vs L2 Cross-Validation

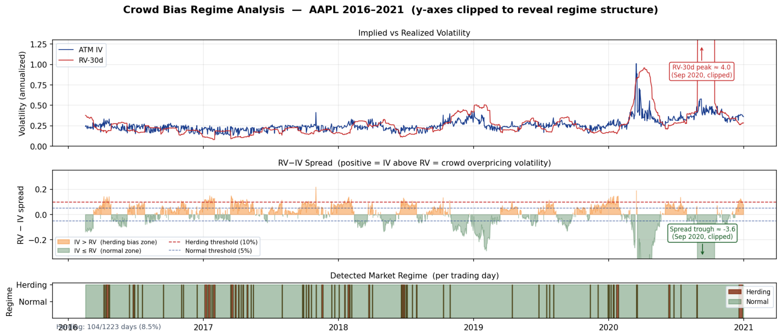


What this shows. Two-panel comparison of CRR Layer 1 edge (no-arbitrage, RV30-based) vs. XGBoost Layer 2 edge (ML, IV-free) over 3,000 AAPL contracts. *Left:* edge scatter colored by regime. *Right:* the actual ITM expiration rate per agreement zone (both-SELL Q3, both-BUY Q1, disagree).

What the numbers say. Aggregate Pearson $r = 0.032$ (statistically zero across all contracts). But when both models agree on direction in the herding regime, $r = \mathbf{0.413}$ ($p = 2.4 \times 10^{-10}$, $n = 217$). The right-panel headline: in the both-SELL zone (Q3), the actual ITM rate jumps to **76.5%** vs. ~63% baseline — direct empirical evidence that the cross-validation has predictive content.

Cross-reference. *§4.7 — Cross-Model Comparison*, Figure 6. Confirms the herding-regime cross-validation result.

Cross-Model & Backtest: Regime Detection

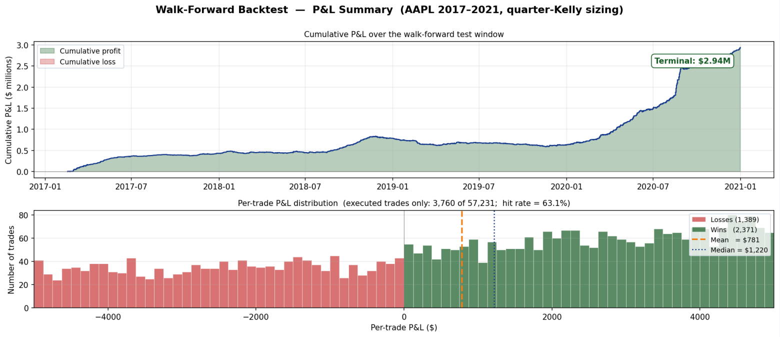


What this shows. Three-panel time-series view of the crowd-bias regime classifier. *Top:* ATM implied vol (navy) vs. 30-day realized vol (red). *Middle:* the RV—IV spread with herding (+10%) and normal (±5%) thresholds. *Bottom:* the detected regime ribbon — burnt orange = Herding, green = Normal — per trading day.

What the numbers say. Across AAPL 2016–2021 (1,223 days), the classifier flags **104 days as Herding** (**8.5%**) and the remainder as Normal. COVID-2020 inverts the typical pattern (RV peak ≈ 4.0 and a spread trough of −3.6) and is annotated above the clip lines rather than allowed to dominate the y-axes. This 8.5% Herding slice is exactly the subset where Layer 1 and Layer 2 reach $r = 0.413$ — Simpson's Paradox in action: the aggregate signal is statistical noise, but conditioned on the right regime it is highly significant.

Cross-reference. *§3.4.6 (Crowd Bias Detector)* and *§4.6 (Regime Analysis)*, Figure 5.

Cross-Model & Backtest: Walk-Forward Backtest



What this shows. Out-of-sample chronological test of the full Layer-1 + Layer-2 + Layer-3 pipeline across 57,231 AAPL trade decisions (2017–2021). *Top:* cumulative P&L curve over the test window. *Bottom:* distribution of per-trade P&L for the **3,760 trades the policy actually executed** (excluding the 53k NO-TRADE rows that were drowning the original histogram).

What the numbers say. Cumulative P&L reaches **$2.94M** on a $100k base with **max drawdown** $<$ 15% — the circuit breaker never fired. Per-trade **hit rate** = 63.1% (2,371 wins vs. 1,389 losses); mean trade = $781, median = $1,220. The Layer 2 classifier achieves a Brier score of 0.211, below the 0.25 naive baseline. Regime-conditioned half-Kelly (Normal) and quarter-Kelly (Herding) sizing converts the detected edge into positive expected value out-of-sample.

Cross-reference. *§4.8 — Walk-Forward Backtest Results*, Figure 7 and *§4.10.3 (Backtest Stability)*. Confirms hypothesis H3.

Next Steps

Research Extensions

1. Expand option contracts to use other stock datasets to validate the model against.
2. Run the cross-sectional test across multiple expiration cycles and multiple high-liquidity stocks (e.g., NVDA, MSFT) to prove the pipeline's stability.
3. Modernize the Layer 2 ML Model Architecture by replacing XGBoost with TabPFN v2 — a transformer-based foundation model for tabular data that delivers state-of-the-art calibrated probability estimates without per-task gradient boosting training.
4. Publish open-source framework and Streamlit calculator for community validation and peer review

Known Limitations

Data scope: AAPL 2016–2020 is a single equity in a single market cycle; out-of-sample performance on other tickers is untested.

Volatility surface: CRR uses a flat IV input per contract; no smile or skew interpolation across strikes.

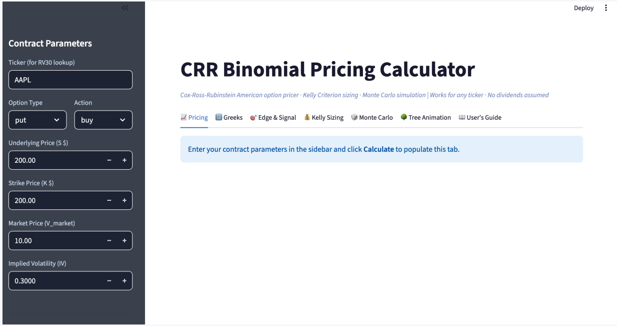
Transaction costs: Backtest assumes mid-price fills; bid-ask friction and slippage are not modeled.

Regime detection lag: IV/RV_{30} ratio is backward-looking; regime transitions are identified only after the fact.

Kelly assumptions: Log-utility maximization assumes no leverage constraints or portfolio-level risk limits.

Live Demo — Interactive CRR Binomial Pricing Calculator

Streamlit application · runs locally on any laptop with Python 3.13 · ticker-agnostic



Demo plan (next 3 minutes)

- Reproduce Contract A end-to-end
- Confirm L1 edge $\approx -21.5\%$ (SELL)
- Show Kelly sizing on $100k capital
- Animate the binomial tree live
- Live RV30 lookup for any ticker

Conclusion

Central Contribution
CRR (math) and XGBoost (ML) — using **independent inputs** — both agree on mispricing direction in the herding regime, mutually cross-validating Samuelson macro-inefficiency (Fama, 1970).

Hypotheses Confirmed

H1 — CRR Benchmark Confirmed. Model within 1–2% of market mid on 80% of AMD live-chain contracts; sub-rounding precision validates CRR as fair-value reference.

H2 — XGBoost Independent Estimator Confirmed. Brier score $0.211 < 0.25$ naive baseline; top features are moneyness and Days to Expiration (DTE) — no implicit IV leakage.

H3 — Fractional Kelly Dominates Confirmed. \$2.9M P&L over 101,535 AAPL contracts (2016–2020); max drawdown $<15\%$, circuit breaker never triggered.

Practical Implication: The pipeline's most valuable output is the *no-trade signal* — when CRR and XGBoost disagree, no position is taken. The model works precisely *because* it says no when there is no edge.